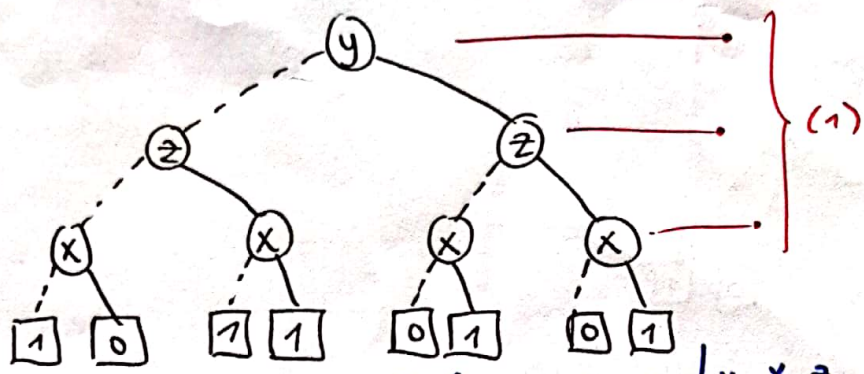
1

a) $\phi = (x \leftrightarrow y) \vee (z \wedge x)$ $\quad y, z, x$



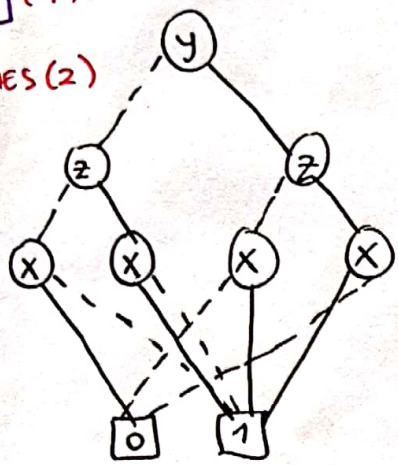
LEAFS: $[0,5] = (\emptyset)$

$[6,7] = [0.5]$

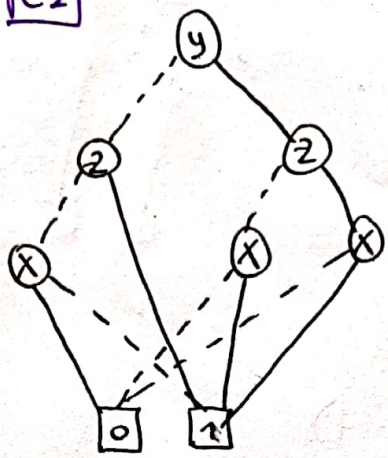
$[8] = (1)$

x,y,z	1	1	1	0	0	1	1	1	y,x,z	1	1	0	1	0	0	1	1
b) x,z,y	1	0	1	0	0	1	1	1								

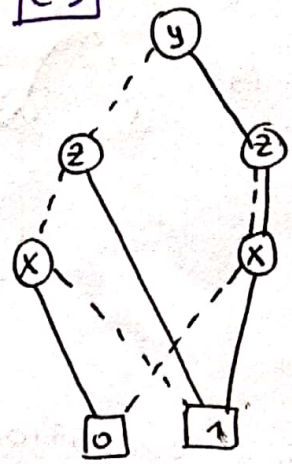
$\boxed{C_1}$ (1) → nodes (2)



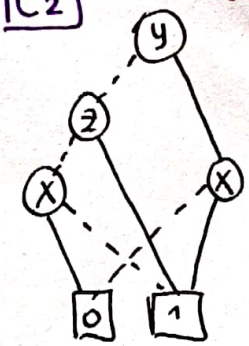
$\boxed{C_2}$ (1)



$\boxed{C_3}$ (1)



$\boxed{C_2}$ (1)



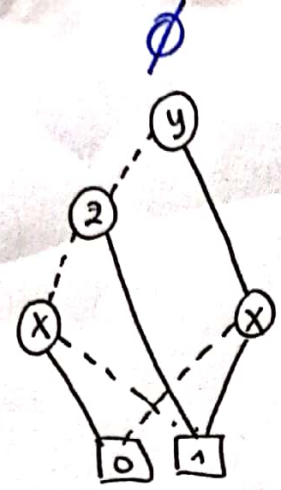
c) SAME WAY AS CONSTRUCTING OBDD FOR $O_1 \vee O_2$. ONLY ↑CE IS IF O_1 OR O_2 IS A LEAF: (1)

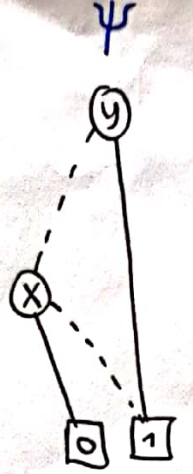
↳ O_1 LEAF
- ↳ $\boxed{0}$ ⇒ 1
- ↳ $\boxed{1}$ ⇒ O_2

↳ O_2 LEAF
- ↳ $\boxed{0}$ ⇒ $\neg O_1$
- ↳ $\boxed{1}$ ⇒ 1

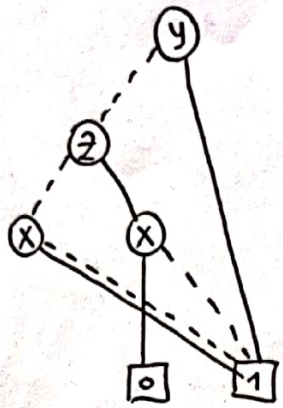
(1)

d)

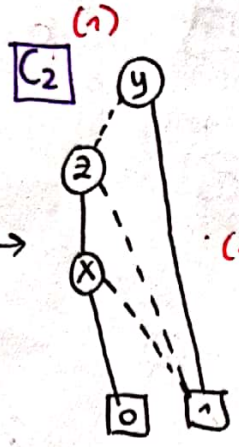




(4)



(1)



(1)

NO LOGICAL OPERATION (-4)
(BUT OBDD)

CONFUSION C_2 & C_3 (-1.5)
(NAMES)

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c) $\equiv$ (1) $\neg O_1$ (-1)
(NEGATION)

NO MENTION OF LEAFS (-1)

CASE 2 (-0.5)

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b) X CORRECT ANS. (-2)